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# Classification Tree

| | Whether | windy | Play Tennis (can) |
|---|---|---|---|
| $S_1$ | Sunny | No | No |
| $S_2$ | Sunny | yes | No |
| $S_3$ | Overcast | No | Yes |
| $S_4$ | Rainy | No | Yes |
| $S_5$ | Rainy | Yes | No |
| $S_6$ | Rain | No | Yes |
| $S_7$ | Overcast | Yes | Yes |
| $S_8$ | Sunny | No | No |

Step 1: Compute root entropy.

$Yes = 4$, $No = 4$, $P(Yes) = \frac{4}{8} = 0.5$

$P(No) = \frac{4}{8} = 0.5$

Entropy, H (measure of impurity in data)
↳ shld be low → better decision - less randomness

$$H_{root} = \sum_{i=1}^{r} -P_i \log_2 P_i$$ → i = no. of classes

for eg - $\sum_{i=1}^{2} P_i \log_2 P_i$

$$= -P_{yes} \log_2 P_{yes} + (-P_{no} \log_2 P_{no})$$

for 2 classes yes & no

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$= -0.5 \log_2(0.5) + (-0.5 \log_2(0.5))$

$\boxed{H_{root} = 1}$

step 2: compute info. gain (IG) for a feature here whether
→ max inform. gain is checked.

whether has 3 values
sunny, Overcast, rain

→ Sunny.
↳ 3 samples → $S_1, S_2, S_8$, (No, No, No) target values.

$P_{No} = \frac{3}{3} = 1$.

only 1 class.

$H_{sunny} = \sum_{i=1}^{1} -P_i \log_2 P_i$

$= -1 \log_2 1$

$H_{sunny} = -1 \times 0 = 0$

entropy is zero → 0 impurity → very pure → or very homogenous.

→ Overcast
↳ 2 samples → $S_3$ & $S_7$ (Yes, Yes)

$P_{yes} = \frac{2}{2} = 1$.

$H_{overcast} = 0$

Rain - 3 samples - $S_4, S_5, S_6$ (Yes, No, Yes)

$P_{yes} = \frac{2}{3}$, $P_{no} = \frac{1}{3}$

$$\sum_{i=1}^{2} -P_i \log_2 P_i$$

$$= -P_{yes} \log_2 P_{yes} + (-P_{no} \log_2 P_{no})$$

$$= -\frac{2}{3} \times -0.585 + \left(-\frac{1}{3} \times -1.585\right)$$

$$= 0.39 + 0.52$$

$$= 0.918$$

Step 3: Compute weighted entropy.

$$H_{weighted} = \frac{\text{No. of samples}}{\text{Total samples}} \times H_{child}$$

$$= \underbrace{\frac{3}{8} \times 0}_{\text{sunny}} + \underbrace{\frac{2}{8} \times 0}_{\text{overcast}} + \underbrace{\frac{3}{8} \times 0.918}_{\text{rain}}$$

$$= 0.344$$

Information gain $(IG)_{weather} = H_{parent} - H_{child\ (weighted)}$

$$= 1 - 0.344$$

$$= 0.656$$